RDIMS 613801

ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Maritime Surface – Sub Surface Officer (MARS - SS) (00230-2)

USQ - Submarine Dived Navigator– LT(N) (ALII~~TBD~~)

MITE Course IDs:
Residential Phase – ~~123456~~121087
OJPR – NA

Formatted: Font: 10 pt, Not Bold

Formatted: Left

Formatted: Not Highlight

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer~~er~~ of the Naval Training Development Center (Atlantic) (NTDC(A)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0		CCSF- Capt(N) Clarke

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

TABLE OF CONTENTS

Title		Page
FOREWORD		5
CHAPTER 1 : TRAINING MANAGEMENT DETAILS		
	Aim	7
	Course Duration	7
	Resources	8
	Course Capacity	8
	MITE (Loading, Scheduling, Prerequisites, Qualification)	8
	Course Reporting	9
	Evaluation	9
	Validation	9
	Qualification	9
	Prior Learning Assessment And Recognition (PLAR)	10
	Special Instructions	10
CHAPTER 2 : PERFORMANCE OBJECTIVES		
PO 001	Plan dived navigation passages	12
PO 002	Navigate a dived submarine	13
CHAPTER 3 : ASSESSMENT OF TRAINEE		
	General	16
	Enabling and Performance Checks	16
	Responsibility For Testing	16
	Progress Monitoring	16
	Training Review Board	17
	Course Reports	17
CHAPTER 4 : ENABLING OBJECTIVES		
EO 001.01	Prepare dived charts	19
EC 001.01		24

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

EO 001.02	Operate ECPINS for dived operations	26
EC 001.02		32
PC 001		34
EO 002.01	Estimate submarine position	36
EC 002.01		39
EO 002.02	Fix a submarine position	40
EC 002.02		45
PC 002		45
ANNEX A	MAIN REFERENCES	1
ANNEX B	TASK LIST	2
ANNEX C	QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS	3
ANNEX D	QSPWB SCALAR	6
ANNEX E	ON JOB PERFORMANCE RECORD (OJPR)	8
ANNEX F	ACRONYMS AND ABBREVIATIONS	19
ANNEX G	SUGGESTED SEQUENCING AND TIMINGS	10
ANNEX H	ADDITIONAL TRAINING RESOURCES	11
ANNEX I	COSTING INFORMATION	12
ANNEX J	COURSE HISTORY	13
ANNEX K	ASSESSMENT PLAN	14
ANNEX L		

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-003/PQ-001 Submarine Dived Navigator Specialty Specification dated 24 01 17~~DD Month YY~~; and

f. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to NTDC(A), who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS (A)		121087	ALII
NFS(P)		121087	ALII

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE APPROVED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Submarine Dived Navigator. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a period of instruction is 50 minutes in duration.

Allocation of ATT	**Periods**
(Sum of all EOs/POs)	24.0
Enabling Checks (ECs)	8.0
Performance Checks (PCs) See Note A.	48.0
Actual Training Time (ATT) Sub-total	80.0 11.4 **trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	1.0
Training Administrative Time (TAT)	2.0
Training Travel Time (TTT)	0
Physical Fitness (PT)	6
Sub-total	9.0 1.2 **trg days**
	0.0
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **7 (50 min)periods** * **Partial days must be rounded up**	89.0 12.7 **trg days**

Note A: PC002 time allocation is 42 periods based on Optimum load of 6 students. Increases and decreases in course load will lengthen or shorten the course by 7 periods / student.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Instructor	LT(N)	Submarine Dived Navigator	1	PO001/002
Assessor	LT(N)	Submarine Dived Navigator	1	PO002
Assessor	LT(N)	Submarine OOW	1	PO002

Critical Training Resources	No of Days
Submarine Trainer for PO, or	6
Operational Submarine	6
Note: 6 days is based on the optimum load	

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS.

a. Maximum – 10;

b. Minimum – 2; and

c. Optimum – 6.

Note: When determining Max, Min and Capacity, the QSPWB is to consult with the TE(S) regarding number of students per course which is based on resource limitations (money, time, instructors to teach and supervise)), equipment limitations (boats, tools, workstations), classroom limitations (desks and chairs), safety concerns (RSO, and Range coaches) and instructional techniques (syndicate size). Operational considerations can be captured if required under Optimum Capacity (4-man attack team, bridge team, RAS team).

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. ~~Course sessions for Naval Reserve Force may be scheduled during training system holidays.~~

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. Minimum Level II security Clearance;

b. Basic Submarine Qualification (AILS);

c. Fleet Navigation Officer (AEEA) or Minor Warship Navigation Officer (MITE 120308); and/or

d. Submarine Warfare Director (AILT).

~~8.~~ Qualification. Successful completion of this course constitutes the primary means of attaining the Submarine Dived ~~Navigator (~~Navigator (ALI~~National Qualification(s)~~). Additionally, the following MITE Course IDs will be granted:

~~a.~~ Submarine Dived Navigator~~Course name~~ - 121087~~MITE Course ID; and~~

~~b.~~8. ~~Course name - MITE Course ID.~~

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years IAW the guidance provided in CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

Formatted: Indent: Left: 1.27 cm, Hanging: 1.27 cm

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Table Numbering, No bullets or numbering, Tab stops: 1.27 cm, List tab

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. Nil.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE APPROVED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 001

1. Performance. Plan dived navigation passages (T0001)

2. Conditions.

   a. Given:

      (1) An equipped and manned Control Room;
      (2) NAV DDS;
      (3) Publications;
      (4) Supervision; and
      (5) Inshore mission.

   b. Denied:

      (1) Nil; and

   c. Environmental:

      (1) Onboard an HMC Submarine (surface/dived);
      (2) Day/Night;
      (3) Any visibility conditions; and
      (4) Any traffic density.

3. Standard. The submarine navigation officer shall plan a dived passage in coastal waters to support an inshore mission IAW refs , by:

   a. Preparing dived charts (T0007);

   b. Configuring ECPINS for dived operations (T0006);

   c. Contributing navigating considerations to mission plan; and

   d. Inputting Mission Events into ECPINS.

4. References. A1, A2, A3, B2, B3

5. Task Numbers. T0001, T0005, T0006.

6. Training Limitations.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 002

1. Performance. Navigate a Dived Submarine (T0002)

2. Conditions.

a. Given:

(1) An equipped and manned Control Room;
(2) NAV DDS;
(3) Publications;
(4) Supervision; and
(5) Inshore mission

b. Denied:

(1) Nil; and

c. Environmental:

(1) Onboard an HMC Submarine (surface/dived);
(2) Day/Night;
(3) Any visibility conditions; and
(4) Any traffic density.

3. Standard. The submarine navigation officer shall navigate a dived passage in coastal waters to support an inshore mission IAW refs , by:

a. Reducing the Pool of Errors, using techniques of;

(1) Depth area inclusion / exclusion;

(2) Single line of bearing;

(3) Range; and

(4) Single line of sounding; ~~and~~

b. Fixing a submarines position through use of bottom contour navigation techniques:

(1) Line of sounding; and

(2) Contour Advancement; ~~and~~

c. Keeping the submarine SAFE (IAW CO Orders); and

d. Managing a mission event timeline +/- 3 minutes.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

4. References. A1, A2, A4, B3, B4

5. Task Numbers. T0002, T0003, T0004

6. Training Limitations. Access to Victoria Submarine Command Team Trainer

Formatted: No underline

Formatted: No underline

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE APPROVED BY COMMANDER OF TRAINING DEVELOPMENT CENTRE ATLANTIC (TDCA)

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of PRACTICAL examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. The NTDC designs all PCs and quality control is maintained at the NFS while administering and scoring Performance Checks. Where possible, PCs should be administered ~~Standards provides quality control by designing, administering and scoring all Performance Checks. Any PC not administered~~ or invigilated ~~by standards staff will be administered~~ by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional and standards staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB or PRB (if required); and

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, 4500-51 - Course Reporting and Certificates.

12. Trainee will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

13. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and / or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE APPROVED BY THE COMMANDER OF TRAINING DEVELOPMENT CENTRE ATLANTIC (TDCA)

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 001 Plan dived navigation passages

EO 001.01

1. Performance. Prepare Dived Charts (T0005) “Diving the Charts”

2. Conditions

   a. Given:

      (1) Inshore mission requirements;
      (2) Electronic charts;
      (3) ECPINS/Laptop; and
      (4) Publications (unclassified/classified;

   b. Denied:

      (1) Nil; and

   c. Environmental:

      (1) Ashore/onboard.

3. Standard. The trainee shall “dive the charts”, IAW mission requirements and refs, by:

   a. Plotting Water Space Management (WSM);

   b. Plotting Prevention of Mutual Interference (PMI);

   c. Plotting physical hazards to dived navigation;

   d. Plotting “no go” lines;

   e. Plotting periscope only depth;

   f. Plotting maximum duck depths;

   g. Plotting safe depth; and

   h. Constructing echo sounder matrix.

4. Time Required

   a. EO – ~~7~~ 6 x 50 min period (~~320~~ 300 Min);

   b. EC – 2 x 50 min period; and

   c. PC – 6 x 50 min period.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

5. Training Aids.

a. ECPINS; and

b. Laptop

6. Test Details.

a. EO 001.01 will be tested in EC 001.01 and PC 001.

7. References.

a. CFCD 110

8. Remarks.

9. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
1. Dived Navigation Theory	1.1 Describe differences between inshore and offshore dived navigation	25			Self Study Guided Discussion	A1, A2, B2	
	2.1 Discuss WSM (K0009)	25			Self Study Guided Discussion	B3	
	2.2 Discuss PMI (K0010)	25			Self Study Guided Discussion	B3	
	2.3 Discuss Hazards to dived	20			Self Study Guided Discussion	A1, A2,	

Formatted Table

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Teaching Point	Sub teaching	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the
	navigation (K0001)						
2. Dived Contour Lines	2.4 Discuss "No Go" line	5			Self Study Guided Discussion	A1, A2	
	2.5 Discuss Periscope only depth	5			Self Study Guided Discussion	A1, A2	
	2.6 Discuss Maximum Duck Depths	5			Self Study Guided Discussion	A1, A2	
	2.7 Discuss Safe Depth	5			Self Study Guided Discussion	A1, A2	
3 Echo sounder matrix	3.1 determine Yellow and Red sounding	10			Self Study Guided Discussion	A1, A3	
	3.2 determine reporting policy	10			Self Study Guided Discussion	A1, A3	
		10					
4. Introduce Mission requirements		50			Discussion /Demo	A4	
5. Dive the Charts				50	Practice	A1, A2	
6. Construct echo sounder matrix				50	Practice	A1, A3	

Formatted Table

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Teaching Point	Sub teaching	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the

Formatted Table

Formatted: Right

~~Instructions: (can be deleted from final version)~~

~~1. All time to be calculated in minutes~~

~~2. Sub Teaching Point includes the verb, what he students learning outcome is based n Blooms cognitive/psycho motor domain level.~~

~~3. Instructional technique – T=Theory, D= Demonstration, and P=Practical~~

~~4. Delivery Approach – F2F = Face to Face, OJT= On the Job training, DL= Distributed Learning, PIP = Pre Instruction Package, and Blended (Time if possible or its not accounted for.~~

~~5. Reference – for the TP or Sub TP down to page and para.~~

~~6. Comments – in relation to the instructional strategy, implication of the intended delivery technique, and or, the ratio of students to instructor for the practical.~~

~~Instructional Methods:~~

~~a. Interactive lecture;~~

~~b. Demonstration – performance;~~

~~c. Guided discussion;~~

~~d. Study assignment;~~

~~e. Self study;~~

~~f. Simulation;~~

~~g. Role play;~~

~~h. Case study;~~

~~i. Tutorial;~~

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

j. Field trip;

k. OJT;

l. Self-paced CBT (asynchronous);

m. Virtual classroom (synchronous); and

n.a. Video teleconference.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Enabling check - EO 001.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Classroom / Submarine.

Arrangements and Procedures

2. EC 001.01 will be a practical / written assessment based on content from EO 001.01.

3. Pass / Fail or Pass mark will be #%.

4. Time required: 2x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 001.01.

2. During this practical assessment, you will be required to dive the charts. You must pass this test to pass EO 001.01. To pass the test, you must complete the performance statement in accordance with the standard.

3. References will be provided.

4. Pass / Fail or Pass mark is #%.

5. You will have 2 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PART IV – ENABLING CHECK LIST

Check List	*SAT*	*UNSAT*	*Comments*

The trainee must receive # SAT to pass this EC.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 001 Plan dived navigation passages

EO 001.02

1. Performance. Operate ECPINS for dived operations

2. Conditions

a. Given:

(1) Inshore mission requirements;
(2) Electronic charts;
(3) ECPINS/Laptop; and
(4) Publications (unclassified/classified;

b. Denied:

(1) Nil; and

c. Environmental:

(1) Ashore/onboard.

3. Standard. The trainee shall prepare ECPINS for dived operations, IAW mission requirements and refs, by:

a. Inputting sensor errors into ECPINS);

b. Inputting Pool of Errors (POE) into ECPINS;

c. Inputting inshore mission events into ECPINS; and

d. Developing visual/electronic fixing schedule.

4. Time Required

a. EO - ~~10~~ 8 x 50 min period (~~500~~ 400 min)

b. EC – 2 x 50 min period; and

c. PC – 6 x 50 min period.

5. Training Aids.

a. ECPINS;

b. Laptop

6. Test Details.

a. EO 001.02 will be tested in EC 001.02 and PC 001.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

7. References.

a. CFCD 110

8. Remarks.

9. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
1. Inertial Navigation Equipment (K0009)	1.1 Describe inertial navigation theory	25 min			Self Study Guided Discussion	A1, B1, B2	
	1.2 Identify inertial navigation equipment	10 min			Self Study Guided Discussion	A1	
	1.3 Discuss Ring laser gyro operation	25 min			Self Study Guided Discussion	A1, B1, B2	
	1.4 Discuss position sources	10 min			Self Study Guided Discussion	A1, B1, B2	
	1.5 Discuss speed log operation	10 min			Self Study Guided Discussion	A1, B1, B2	
2. Sensor Errors	2.1 Identify sensor errors types (K0011)	50 min			Self Study Guided Discussion	A1, B1, B2	

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
	2.2 Evaluate Navigation Sensor information	10 min		15 min	Self Study Guided Discussion /practice	A1, B1, B2	
3. Pool of Errors	3.1 Discuss Pool of Errors theory	30 min			Self Study Guided Discussion	A1, B1	
	3.2 Discuss impact of POE on MLA	10 min			Self Study Guided Discussion	A1, B1	
4. Calculate Pool of Errors (POE)	4.1 Fix errors	10 min		10 min	Self Study Guided Discussion /Practice	A1	
	4.2 Distance/ speed log error	10 min		10 min	Self Study Guided Discussion /Practice	A1	
5. Interpret set and drift	5.1 Tidal Stream and current prediction	10 min		10 min	Self Study Guided Discussion /Practice	A1, B1	
	5.2 sub surface tidal streams ,	10 min		10 min	Self Study Guided Discussion /Practice	A1, B1	
	5.3 sub surface ocean	10 min		10 min	Self Study Guided Discussion	A1, B1	

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

	currents				/Practice		
6. set and drift errors ellipse or circular	6.1 Error ellipse for rectilinear flows	10 min		10 min	Self Study Guided Discussion /Practice	A1, B1	
	6.2 Circular error for rotary flow	10 min		10 min	Self Study Guided Discussion /Practice	A1, B1	
7. Input pool of errors into ECPINS	7.1 Select Pool of Errors in ECPINS Menu	10 min	15 min	25 min	Self Study Guided Discussion /Demo/ Practice	A1	
8. Submarine Handling	8.1 Discuss Submarine Handling (Classified)	50 min			Self Study Guided Discussion		
9. Contribute Navigation considerations to Plans	9.1 Develop a fixing plan to Where When Method	20 min			Self Study Guided Discussion		
	9.2 Extract mission events from ops plan	10 min		15 min	Self Study Guided Discussion /Practice		
	9.3 Plan periscope photograph events	25 min		10 min	Self Study Guided Discussion /Practice		
	9.4 Input inshore mission events			15 min	Self Study Guided Discussion /Practice		

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

	into ECPINS						

~~Instructions: (can be deleted from final version)~~

~~1. All time to be calculated in minutes~~

~~2. Sub Teaching Point- includes the verb, what he students learning outcome is based n Blooms cognitive/psycho motor domain level.~~

~~3. Instructional technique – T=Theory, D= Demonstration, and P=Practical~~

~~4. Delivery Approach – F2F = Face to Face, OJT= On the Job training, DL= Distributed Learning, PIP = Pre Instruction Package, and Blended (Time if possible or its not accounted for.~~

~~5. Reference – for the TP or Sub TP down to page and para.~~

~~6. Comments – in relation to the instructional strategy, implication of the intended delivery technique, and or, the ratio of students to instructor for the practical.~~

~~Instructional Methods:~~

- ~~o. Interactive lecture;~~
- ~~p. Demonstration - performance;~~
- ~~q. Guided discussion;~~
- ~~r. Study assignment;~~
- ~~s. Self-study;~~
- ~~t. Simulation;~~
- ~~u. Role play;~~
- ~~v. Case study;~~
- ~~w. Tutorial;~~
- ~~x. Field trip;~~

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

~~y. OJT;~~

~~z. Self-paced CBT (asynchronous);~~

~~aa. Virtual classroom (synchronous); and~~

~~bb. Video teleconference~~

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Enabling check - EO 001.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Classroom / Submarine.

Arrangements and Procedures

2. EC 001.02 will be a practical / written assessment based on content from EO 001.02.

3. Pass / Fail or Pass mark will be #%.

4. Time required: 2 x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 001.02 will be a practical assessment based on content from EO 001.02.

7. During this practical assessment, you will be required to prepare ECPINS for dived operations. You must pass this test to pass EO 001.02. To pass the test, you must complete the performance statement in accordance with the standard.

8. References will be provided.

9. Pass / Fail or Pass mark is #%.

10. You will have 2x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PART IV – ENABLING CHECK LIST

Check List	*SAT*	*UNSAT*	*Comments*

The trainee must receive # SAT to pass this EC.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PERFORMANCE CHECK - PO 001

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Classroom / Ship / Submarine.

Arrangements and Procedures

2. PC 001 will be a practical / test based on content from PO 001.

3. A rating of SAT musty be achieved in all categories.~~Pass / Fail or Pass mark will be #%~~.

4. Time required: 6 x 50 min period.

5. References will / will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will be required to plan a dived navigation passage. You must pass this test to pass PO 001. To pass the test, you must complete the performance statement in accordance with the standard.

3. References will be provided.

4. You must achieve SAT in all categories to Pass~~ / Fail or Pass mark is #%~~.

5. You will have 6x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Give guidance for the development of the assessment.

PART IV – PERFORMANCE CHECK LIST

The trainee must receive # SAT to pass this PC.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Dived Passage – Planning

Student: _____ Assessor: _____

Passage: _____

Objective	Grade (UNSAT/NI/SAT/VSAT)	Remarks
Planning		
1. Selection of tracks.		
2. Completeness of chart work.		
3. HOT / Tidal Set / Wind / Current.		
4. Command decision points, safety margins and contours.		
5. Secondary & evasion routes.		
6. Traffic.		
7. Fishing vessel safety considerations.		
8. Battery Considerations.		
9. Shiphandling considerations.		
10. Communications.		
11. Important aspects well presented.		
12. Conduct of passage covered.		
13. Changes to plan incorporated.		

UNSAT / NI / SAT

(In order to achieve the requirements the student must achieve SAT, please circle the appropriate assessment above.)

Assessor's signature: _____ Student's signature: _____

Date: _____

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 002 Navigate a Dived Submarine (T0002)

EO 002.01

1. Performance. Estimate submarine position (T0003)

2. Conditions

   a. Given:

      (1) Inshore mission requirements;
      (2) Electronic charts;
      (3) ECPINS/Laptop; and
      (4) Publications (unclassified/classified;

   b. Denied:

      (1) Nil; and

   c. Environmental:

      (1) Ashore/onboard.

3. Standard. The trainee shall manage a pool of errors by:

   a. Reducing a Pool of Errors (POE);

      (1) Depth area inclusion / exclusion;

      (2) Single line of sounding;

      (3) Range; and

      (4) Single line of bearing; and

   b. Calculating a future POE.

4. Time Required.

   a. EO - 6x 50 min period (280 min)

   b. EC – 2x 50 min period; and

   c. PC – 42 x 50 min period.

5. Training Aids.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

~~c.~~a. ECPINS; and

~~d.~~b. Laptop.

6. Test Details.

b. EO 002.01 will be tested in EC 002.01 and PC 002.

7. References.

a. CFCD110

8. Remarks.

9. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
1. Reduce a POE.	1.1 Depth area inclusion/ exclusion,	10 min		50 min	Discussion /Simulation	A1	
	1.2 Single line sounding	10 min		50 min	Discussion /Simulation	A1	
	1.3 Line of Bearing using Radar, Sonar, and Periscope	10 min		50 min	Discussion /Simulation	A1	
	1.4 Range using Radar, Sonar, and Periscope	10 min		50 min	Discussion /Simulation	A1	
2. Calculate a future POE		20 min		20 min	Discussion /Simulation	A1	

Formatted: Outline numbered + Level: 2 + Numbering Style: a, b, c, … + Start at: 1 + Alignment: Left + Aligned at: 0.63 cm + Tab after: 1.27 cm + Indent at: 1.27 cm

Formatted: Indent: First line: 0 cm, Outline numbered + Level: 2 + Numbering Style: a, b, c, … + Start at: 1 + Alignment: Left + Aligned at: 0.63 cm + Tab after: 1.27 cm + Indent at: 1.27 cm

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

~~Instructions: (can be deleted from final version)~~

~~1. All time to be calculated in minutes~~

~~2. Sub Teaching Point- includes the verb, what he students learning outcome is based n Blooms cognitive/psycho motor domain level.~~

~~3. Instructional technique – T=Theory, D= Demonstration, and P=Practical~~

~~4. Delivery Approach – F2F = Face to Face, OJT= On the Job training, DL= Distributed Learning, PIP = Pre Instruction Package, and Blended (Time if possible or its not accounted for.~~

~~5. Reference – for the TP or Sub TP down to page and para.~~

~~6. Comments – in relation to the instructional strategy, implication of the intended delivery technique, and or, the ratio of students to instructor for the practical.~~

~~Instructional Methods:~~

- ~~cc. Interactive lecture;~~
- ~~dd. Demonstration – performance;~~
- ~~ee. Guided discussion;~~
- ~~ff. Study assignment;~~
- ~~gg. Self study;~~
- ~~hh. Simulation;~~
- ~~ii. Role play;~~
- ~~jj. Case study;~~
- ~~kk. Tutorial;~~
- ~~ll. Field trip;~~
- ~~mm. OJT;~~
- ~~nn. Self-paced CBT (asynchronous);~~
- ~~oo. Virtual classroom (synchronous); and~~
- ~~pp. Video teleconference~~

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Enabling check - EO 002.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Classroom / Submarine.

Arrangements and Procedures

2. EC 002.01 will be a practical assessment based on content from EO 001.02.

3. Pass / Fail or Pass mark will be #%.

4. Time required: 2x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical assessment based on content from EO 002.01.

2. During this practical assessment, you will be required to estimate a submarines position. You must pass this test to pass EO 002.01. To pass the test, you must complete the performance statement in accordance with the standard.

3. References will not be provided.

4. Pass / Fail or Pass mark is #%.

5. You will have 2 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PO 002 Navigate a Dived Submarine (T0002)

EO 002.02

1. Performance. Fix a Submarine position

2. Conditions

   a. Given:

      (1) Inshore mission requirements;
      (2) Electronic charts;
      (3) ECPINS/Laptop; and
      (4) Publications (unclassified/classified);

   b. Denied:

      (1) Nil; and

   c. Environmental:

      (1) Ashore/onboard.

3. Standard. The trainee shall fix the position for a dived submarine, by:

   a. Using Bottom contour fixing techniques (K005);

      (1) Line of soundings;

      (2) Contour advancement; and

   b. Directing the fixing routine.

4. Time Required.

   d. EO – ~~2~~ 7 x 50 min period (~~105~~ 305 min);

   e. EC – 2x 50 min period; and

   f. PC – 42 x 50 min period.

5. Training Aids.

   a. ECPINS; and

   b. Laptop.

6. Test Details.

   a. EO 002.02 will be tested in EC 002.02 and PC 002.

7. References.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

a. CFCD110

8. Remarks.

9. Teaching Points

Teaching Point	Sub teaching Point	Instructional Technique (Time in min)			Delivery Approach	Ref	Comments supporting the Instructional Strategy
		T	D	P			
1. Bottom Contour Fixing	1.1 Describe Technique	25 min			Self Study/Discussion	A1	
2. Fix position using bottom contours	2.1 Line of Sounding	10 min		20 min	Self Study/ Discussion / Simulation	A1	
	2.2 Contour Advancement	10 min			Self Study/Discussion	A1	
3. Direct fixing routine	3.1 Tactical considerations	10 min		10 min	Self Study/ Discussion / Simulation	A1, A2, A4, B3, B4	
	3.2 Frequency and method	10 min		10 min	Self Study/ Discussion / Simulation	A1	
4. Navigation Principles	Navigation Principles	195 min			Case Study		

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Instructions: (can be deleted from final version)

1. All time to be calculated in minutes

2. Sub Teaching Point- includes the verb, what he students learning outcome is based n Blooms cognitive/psycho motor domain level.

3. Instructional technique – T=Theory, D= Demonstration, and P=Practical

4. Delivery Approach – F2F = Face to Face, OJT= On the Job training, DL= Distributed Learning, PIP = Pre Instruction Package, and Blended (Time if possible or its not accounted for.

5. Reference – for the TP or Sub TP down to page and para.

6. Comments – in relation to the instructional strategy, implication of the intended delivery technique, and or, the ratio of students to instructor for the practical.

Instructional Methods:

qq. Interactive lecture;

rr. Demonstration – performance;

ss. Guided discussion;

tt. Study assignment;

uu. Self-study;

vv. Simulation;

ww. Role play;

xx. Case study;

yy. Tutorial;

zz. Field trip;

aaa. OJT;

bbb. Self-paced CBT (asynchronous);

ccc. Virtual classroom (synchronous); and

ddd. b. Video teleconference

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Enabling check - EO 002.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Classroom / Submarine.

Arrangements and Procedures

2. EC 002.01 will be a practical / written assessment based on content from EO 001.02.

3. SAT must be achieved in all categories to achieve Pass.~~Pass / Fail or Pass mark will be #%.~~

4. Time required: 2x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical assessment based on content from EO 002.01.

2. During this practical assessment, you will be required to fix a submarine position. You must pass this test to pass EO 002.01. To pass the test, you must complete the performance statement in accordance with the standard.

3. References will be provided.

4. You must achieve SAT in all categories to Pass.~~Pass / Fail or Pass mark is #%.~~

5. You will have 2 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Check List	*SAT*	*UNSAT*	*Comments*

The trainee must receive # SAT to pass this EC.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

PERFORMANCE CHECK - PO 002

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer / Ship / Submarine.

Note: A sea run is a desirable assessment; however, realizing the difficulty in accessing submarine sea days, CCSF will determine COA for assessment outside of the VSCTT.

Arrangements and Procedures

2. PC 001 will be a practical test based on content from PO 002.

3. Pass / Fail

4. Time required: 42 x 50 min period. Time determination is 7 periods/student at optimum load. Increases and decreases in course load will require the course to lengthen or shorten.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will be required to navigate a dived submarine. You must pass this test to pass PO 002. To pass the test, you must complete the performance statement in accordance with the standard.

3. References will be provided.

4. Pass / Fail

5. You will have 7 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Give guidance for the development of the assessment.

PART IV – PERFORMANCE CHECK LIST

The trainee must receive # all SAT to pass this PC.

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Dived Passage – Brief and Passage Execution

Student: _____ Assessor: _____

Passage: _____

Objective	Grade*	Comments
Brief		
1. Confident and concise.		
2. Comprehensive.		
3. Handling of questions.		

UNSAT / NI / SAT

(In order to achieve the requirements the student must achieve SAT, please circle the appropriate assessment above.)

Assessor's signature: _____ Student's signature: _____

Date: _____

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Dived Passage Assessment

Objective	Grade*	Comments
1. Fixing/Chartwork.		
2. Basic Navigation Skills.		
3. Submarine Dived Navigation technical skills.		
4. Information Management/Evaluation.		
5. Establish priorities.		
6. Control of the team (OOW/2OOW/COTW)		
7. Situational awareness. Focus ahead of events.		
8. Flexibility. Ability to deviate from the plan.		
9. Advice to Command. Information flow.		
10. Contact avoidance IAW CSOs.		
12. Shiphandling.		
13. ETA Management (+/- ____ minutes)		
14. Turnover		

***Note: Grade = UNSAT, NI, SAT**

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Dived Passage Assessment

NavO:	Assessor:	CO:
Transit From:	To:	Date:
DTG: SOA:	Visibility: nm	Day / Night (circle one)
Pilotage difficulty: low / med / high (circle one)	Traffic Level: low / med / high (circle one)	
Type: ECPINS	Method: Visual / Restricted (circle one)	
Developmental VSCTT: ☐ Run # _____ Assessed VSCTT: ☐ Run # _____	Developmental Sea: ☐ Run # _____ Assessed Sea: ☐ Run # _____	

Strengths / Areas for Development:

Developmental Runs

UNSAT / NI / SAT
(circle one)

A complete debrief has been conducted and the trainee has seen the content of this assessment form:

CTO's Signature: _____ Trainee's Signature: _____

Date: _____

Assessed Runs

UNSAT / NI / SAT

(To achieve the requirements of PCL 406A / PCL 406B the trainee must achieve SAT. Please circle the appropriate assessment above.)

A complete debrief has been conducted and the trainee has seen the content of this assessment form:

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

Dived Passage Assessment

CTO's Signature: _____ Trainee's Signature: _____

Date: _____

Formatted: Not Highlight

Formatted: Not Highlight

Formatted: Not Highlight

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	CFCD 130 – Canadian Navigation Manual
A2	CFCD110 – Submarine Operating Instructions
A3	VCEOPs – Victoria Class Emergency Operating Procedures
A4	NCPM 411 – Naval Combat Procedures Victoria Class Standard Tactical Operating Procedures (STOPS)
B1	Admirality Manaual of Navigation – BR 45 Vol 3
B2	USN SSN Navigation Handbook (RLGN 3rd edition)
B3	ATP 18 – Alleid Manual of Submarine Operations
B4	SMP 108 – Upholder Command Tactical Guidance

ANNEX B – TASK LIST

Task No	Task Statement	Trg / No Trg	PO
T0001	Planned dived navigation	Y	001
T0002	Navigate a dived submarine	Y	002
T0003	Estimate submarine position	Y	002
T0004	Manage a pool of errors (POE)	Y	002
T0005	“Diving the Charts”	Y	001
T0006	Operate ECPINS for dived operations	Y	001

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 22 November to 2 December 2016

Location: NPTG HQ Esquimalt and TDC Atlantic

Board Members:

Surname	Initials	Rank	MOSID	Unit
Meadley	J.D.	LT(N)	00211	NPTG HQ
Pischedda	J.P.	LT(N)	00207-2	CCSF
Darsigny	R.D	LT(N)	00207-2	TDCA

Board Objective: This board was convened to consolidate and align the development of the Submarine Dived Navigator QSP which was required as a result of the separation of the Navigation responsibilities from those of the Submarine Warfare Director. A small focused board was designed to complete this task in an expedited manner. Board was required to address the Cornerbrook BOI recommendations and the development of a Submarine Dived Navigator USQ.

Specification Task – No Train Decisions:

Task No	Task Statement	Substantiation
0001	Planned dived navigation	All task in a USQ are trained
0002	Navigate a dived submarine	All task in a USQ are trained
0003	Estimate submarine position	All task in a USQ are trained
0004	Manage a pool of errors (POE)	All task in a USQ are trained
0005	“Diving the Charts”	All task in a USQ are trained
006	Operate ECPINS for dived operations	All task in a USQ are trained

Discussions / Decisions:

1. Informal, and decentralized discussion were the main drivers of the decisions. USQ tasks and knowledges were socialised with CCSF staff which led to two distinct Performance Objectives, one on planning and one on execution.

2. The capability of the Submarine trainer for practice and assessment was of concern especially with the fidelity of the electronic charts and the reaction of the navigation sensors.

3. Realised that much of the fine tuning of the TPs timing, references and methodology will have to be completed by TDCA Staff.

4. TDCA will have to develop formative and summative assessments to support the assessment plan.

Lessons Learned: NIL

Guidance to Instructional Staff:

1. Use of Tel is advised for Theory components of EO001.01. All Practical application can be completed using ECPINS software and a laptop.

2. Use of submarine simulator is recommended for all practice and assessment for PO 002. Recommend that the fidelity of the ECPINS charts loaded be addressed as well as the Navigation inputs to the ECPINS

3. Ensure the terms used in the TPs are aligned with the references

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Add reference details	TDC (A)	20 Jan 17
2	Develop EC's practical	TDC (A)	20 Jan 17
3	Develop course costs	TDC (A)	20 Jan 17
4	Develop sequence of instruction	TDC(A)	20 Jan 17
5	Rationalized Instructional time	TDC (A)	20 Jan 17
6	Review and align assessment plan		
7	TEL analysis	TDC (A)	5 Jun 17
8	Request Pilot course	TDC (A)	20 Jan 17

Other Considerations (if applicable):

SPQR Approval

Approval Date: 24/01/2017 A-PD-055-003/PQ-

OSS ALII

Submarine Dived Navigator

Section 1 - General

1. Military Occupational Structure (MOS)

Applicable to the following components (MOS ID):

Special Force (Mobilization)

00207.3.N.03Maritime Surface and Sub-Surface – Submarine

Regular Force

00207.1.N.02Maritime Surface and Sub-Surface – Submarine

2. Functional Description

Possession of this specialty will enable submarine officers to navigate a submarine while dived in a safe and competent manner.

3. Working Conditions

The working environment encountered includes all conditions applicable to the occupation. In addition, employment in submarines provides a set of working conditions for which there is no equivalent in the surface fleet. Submarine duties involve an extremely restrictive environment where noise, extremes of temperature, high humidity, atmospheric contaminants, fluctuating atmospheric pressure and lack of sleep contribute to high levels of physical strain and fatigue. The requirement to live and work for extended periods in close quarters with a large number of other personnel in cramped, unpleasant conditions creates a high degree of emotional stress, which is exacerbated by the fact that a simple error at any time while the submarine is submerged could have catastrophic consequences for the entire crew. Submarine duties may lead to physical disorders arising from extended periods of exposure to atmospheric contaminants such as high partial pressures of carbon dioxide and significant concentrations of carbon monoxide, hydrocarbons, chlorine, stibine, arsine and the combustion products of Freon. Reduction in sight efficiency may result from extended periods of exposure to inadequate illumination and reduction in hearing efficiency may result from exposure to engine and ventilation system noise and rapid changes in air pressure.

Formatted: English (Canada)

4. Prerequisite(s)

Personnel must meet the following requirements to be selected for this specialty:

a. Minimum Level II security clearance

b. Basic Submarine Qualification (AILS)

c. Fleet Navigation Officer (AEEA) or Minor Warship Navigation Officer (MITE 120308), or

d. Submarine Warfare Director (AILT)

5. Qualification Method(s)

a. Formal Training

Personnel shall be awarded the qualification after successful completion of a residential course, and endorsement from the Commanding Officer.

Section 2 - Performance Requirements

Task

T0001 Plan Dived Navigation

T0002 Navigate a Dived Submarine

T0003 Estimate Submarine Position

T0004 Mange Pool of Errors (POE)

T0005 "Diving the Charts" (define danger lines)

T0006 Operate ECPINS for Dived Operations

Skill

Not Applicable

Knowledge

K0001 Hazards to Dived Navigation

K0002 Dived Navigation Theory

K0003 Dived Navigation Equipment

K0004 Pool of Errors Theory

K0005 Bottom Contour Navigation

K0006 Dived Charts

K0007 Water Space Management (WSM)

Formatted: English (Canada)

K0008 Prevention of Mutual Interference (PIM)

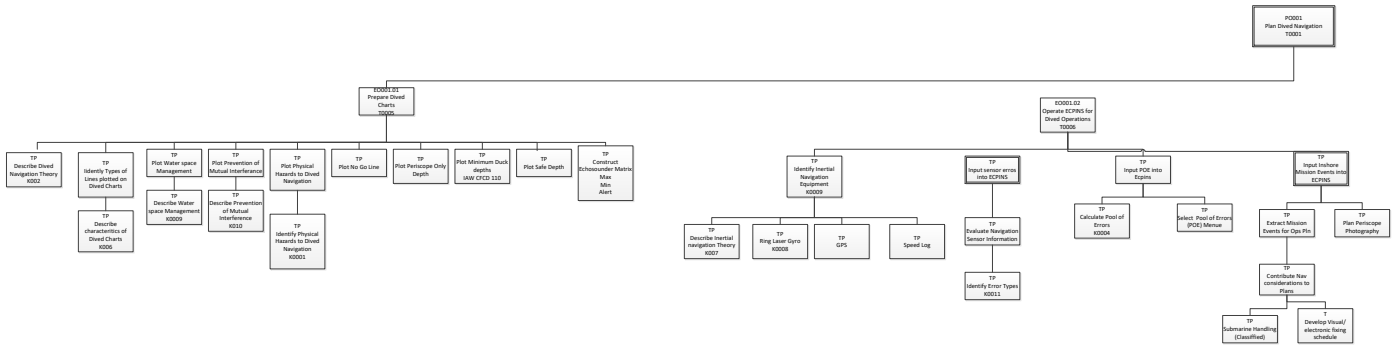
K0009 ECPINS

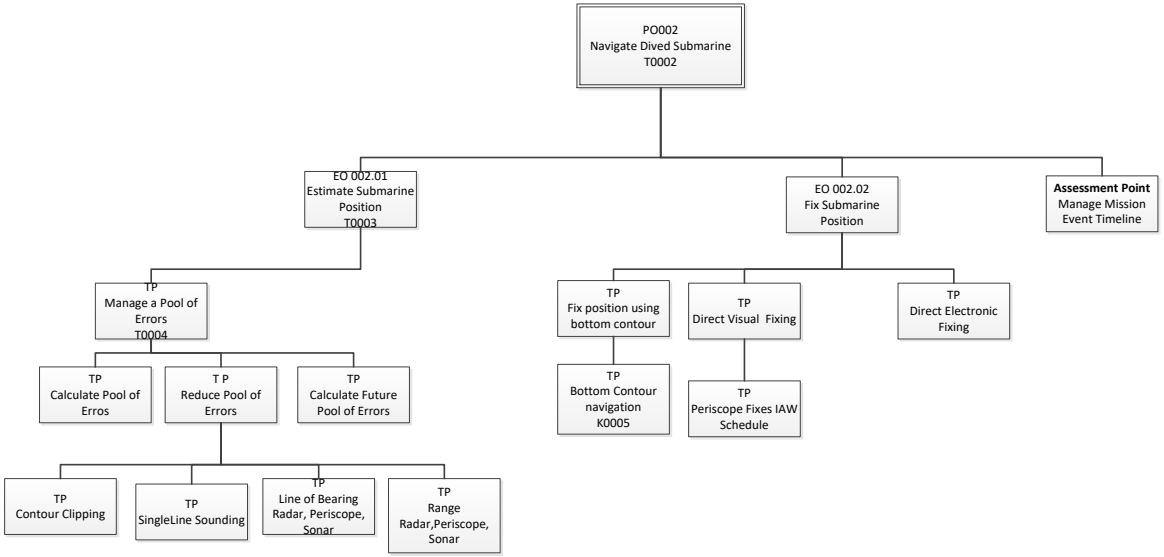
K0010 Inertial Navigation System

Formatted: Space Before: 9 pt, Tab stops: 0.16 cm, Left + Not at 16.4 cm

ANNEX D – QSPWB SCALAR

Formatted: French (Canada)




PO002
Navigate Dived Submarine
T0002
EO 002.01
Estimate Submarine Position
T0003
EO 002.02
Fix Submarine Position
Assessment Point
Manage Mission Event Timeline
TP
Manage a Pool of Errors
T0004
TP
Fix position using bottom contour
TP
Direct Visual Fixing
TP
Direct Electronic Fixing
TP
Calculate Pool of Erros
T P
Reduce Pool of Errors
TP
Calculate Future Pool of Errors
TP
Bottom Contour navigation
K0005
TP
Periscope Fixes IAW Schedule
TP
Contour Clipping
TP
SingleLine Sounding
TP
Line of Bearing
Radar, Periscope, Sonar
TP
Range
Radar,Periscope, Sonar

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

NA

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
VSCTT	Victoria Submarine Command Team Trainer
POE	Pool of Errors
PMI	Prevention of Mutual Interference
WSM	Water Space Management
ECPINS	Electronic Charting and Position Indicating System

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO / EO	PO / EO Title	EO	EC*	PC*	PO
001.01	Prepare Dived charts	6	2		
001.02	Operate ECPINS for dived operations	8	2		
002.01	Estimate Submarine Position	6	2		
002.02	Fix a Submarine Position	6	2		
PC 001	Plan Dived Nav Passages			6 (1)	
PC 002	Navigate a Dived Submarine			42	
	Total				

* Includes review and debrief time.

Note 1: The QSP calls for 6 hours per person to conduct PC 001. Only one day is required in the schedule as each succession student can perform PC 001 while the previous student is performing PC 002.

Formatted: Font: Not Bold

Formatted: Font: Not Bold

Formatted: Font: Not Bold

Formatted: Font: Not Bold

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).

Template Annex I Costing.xls

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	146	$1,000.0		$146,000	$146,000	$7030,000.00
R&Q	146	$36.0	20	$10,0804,320	$10,0804320	$50,400.0021,600.00
Incidentals	146	$17.5	20	$4,9002,100	$4,9002,100	$2410,500.00
Subtotal				**$28,98012,420**	**$12,42028,980**	**$144,900.0062,100**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800300	$800300	$4,0001,500.00
Classroom (square meters)	30	$13.00	1513	$5070…	…$5070	…$25,350
Computer requirements	167	$1,000.0	1	$167,000	$0	$167,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc)	156	$30.0	1	$450180	$450180	$2,250900.00
Subtotal				**$28,26523,565**	**$1,2655565**	**$33,32545,825.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal	-	$8,920	$8,920	$44,600.00
	Total Estimated Costs			**$66,16535,985**	**$39,16517,985**	**$222,825107,925**

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

ANNEX K – ASSESSMENT PLAN

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
EC 001.02	Tps 4,5,6	To prepare the ECPINS charts for a dived inshore passage IAW a mission scenario.	Practical	1. All Lines plotted 2. Construct Echo Sounder Martix	This EC is only a confirmation of the safe operation IAW CCD110
EC 001.02	Tps 4,5,6,7,9	To confirm that a POE can be constructed.	Practical	1. Pool of errors completed taking into account Set and Drift, Tides, Current. 2. Fixing plan is developed	This EC is confirmation that the trainee can build a POE to support PO 002.
PC 001	EO 001.01 EO 001.02	To build their electronic charts, POE and fixing schedule to support and inshore mission.	Practical	IAW checklist	Cannot progress to PC 002 without this being complete
EC 002.01	TPs, 2,3	To confirm that a dived submarine position can be estimated and as a POE increases, techniques to reduce are applied.	Practical	1. Use Depth area inclusion/exclusion 2 Use single line	This EC is confirmation that the trainee can estimate a dived submarines position manage a POE.

				of sounding 3. Use one Line of bearing 4. Range using one sensor	
EC 002.01	TPs, 1, 2	To confirm that fixing through Line of sounding, and contour advancement is used and to direct fixing IAW the routine		1. Use Line of sounding 2 Use Contour Advancement 3.Direct fixing routine	This EC is confirmation that the trainee can manage a POE and predict a future POE
PC 002	~~EO 001.01~~ ~~EO 001.02~~ EO 002.01 EO 002.02	Cumulative assessment where the navigator executes a dived inshore passage in support of a mission, This can be completed on board, at sea or in the submarine simulator.		1. Keeps submarine safe through effective use of the POE. 2. Fixes the submarine position 3. Manages event activities.	This PC is last requirement to achieve the Submarine Dived Navigation.